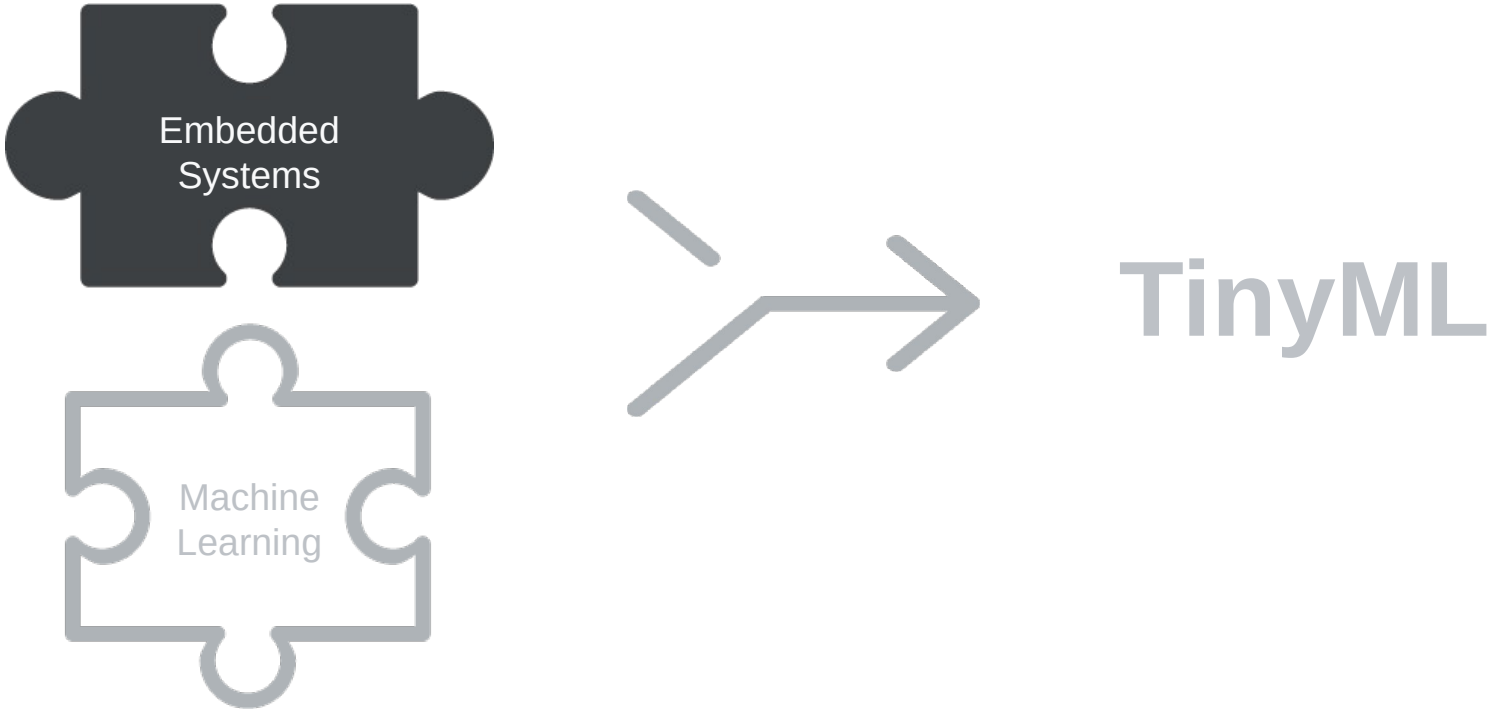
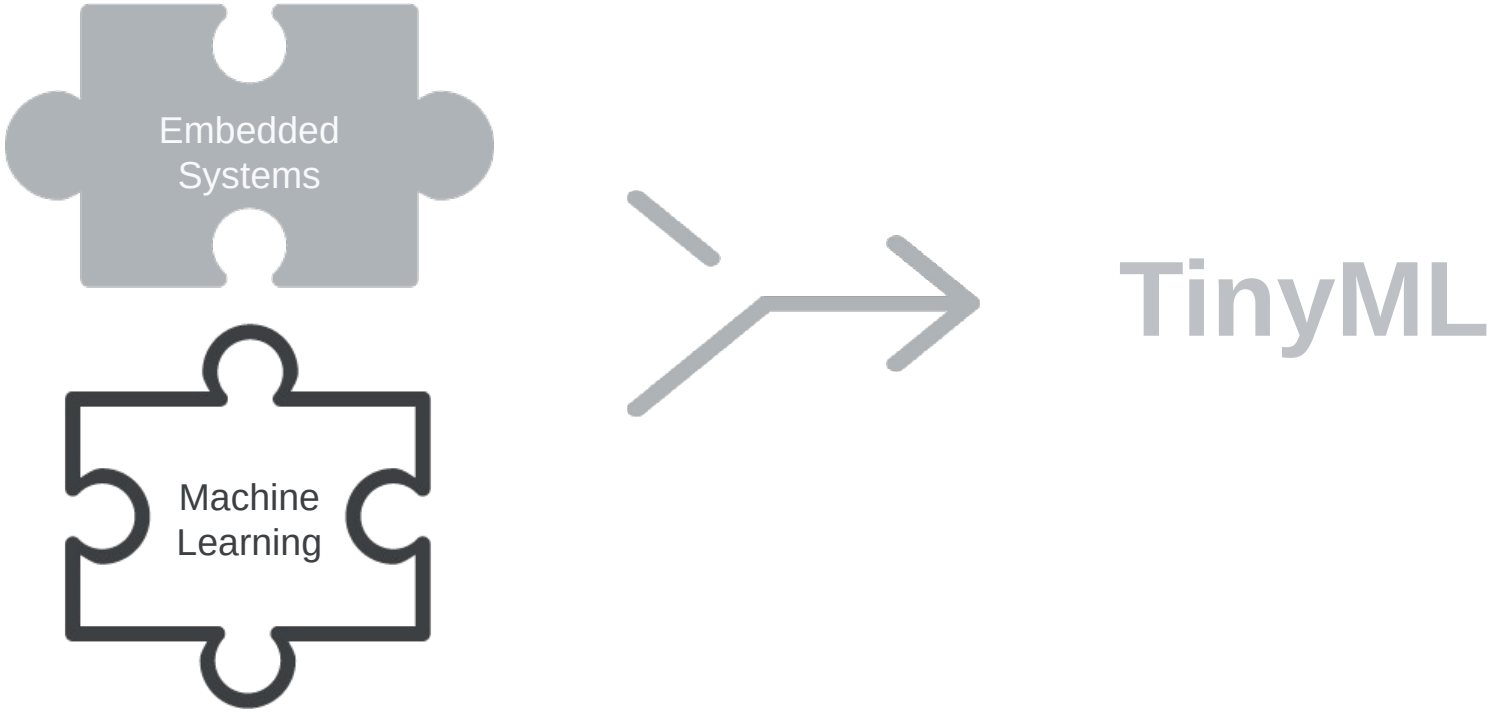
How Do We Enable TinyML?

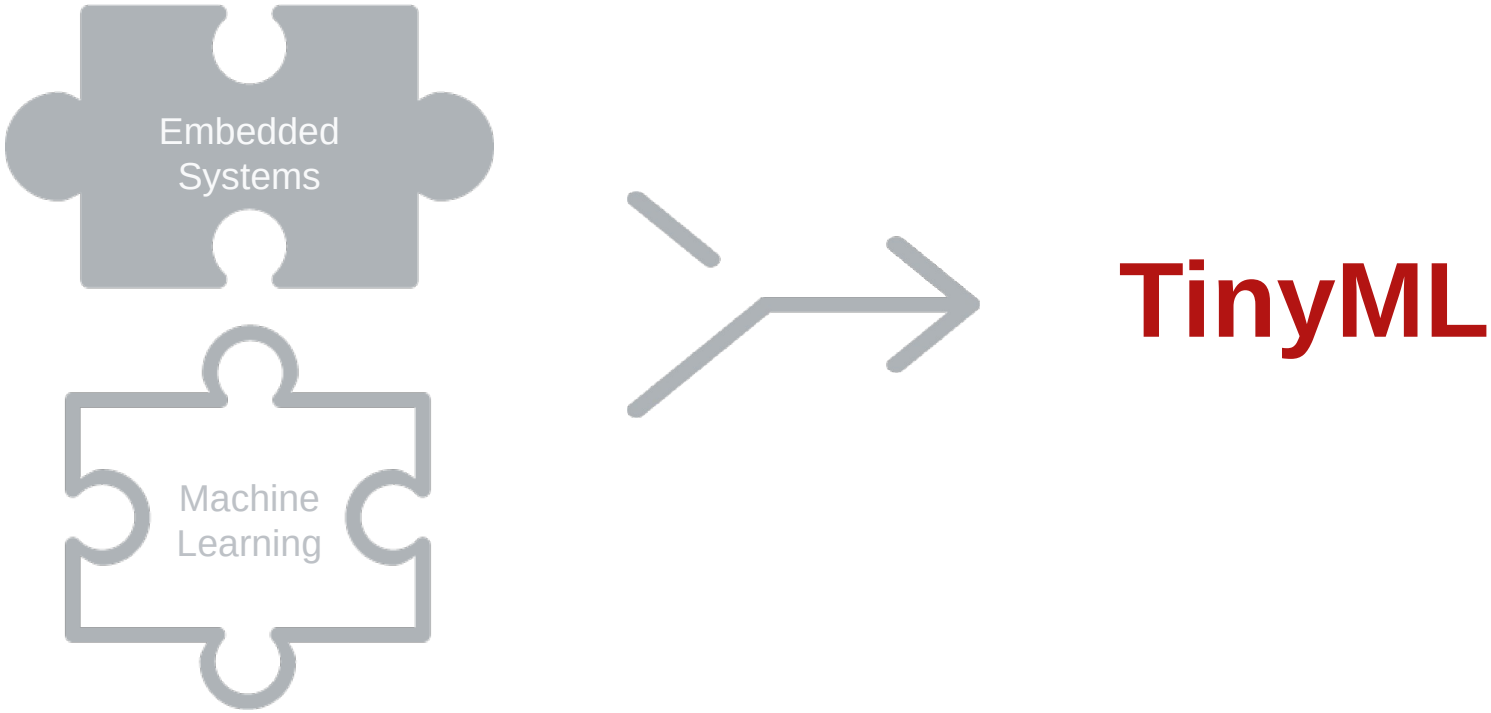
What Makes **TinyML**?



What Makes **TinyML**?



What Makes **TinyML**?



Let’s Take an Example

Let’s Take an Example

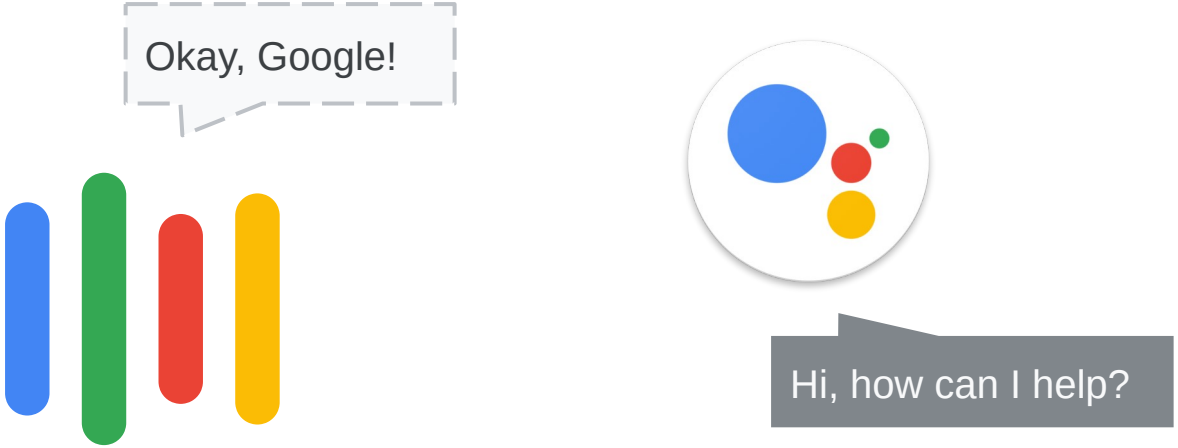






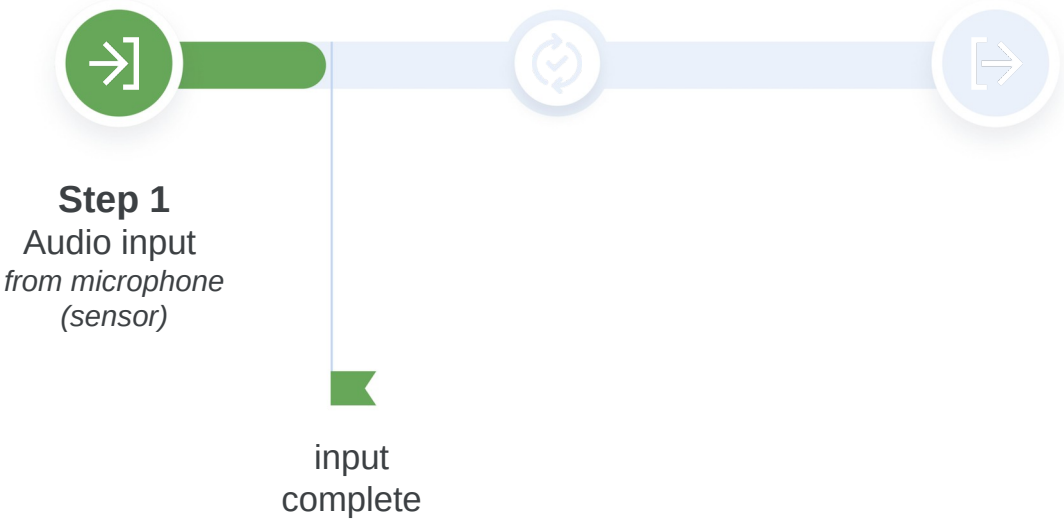
Google Assistant

Let’s Take an Example

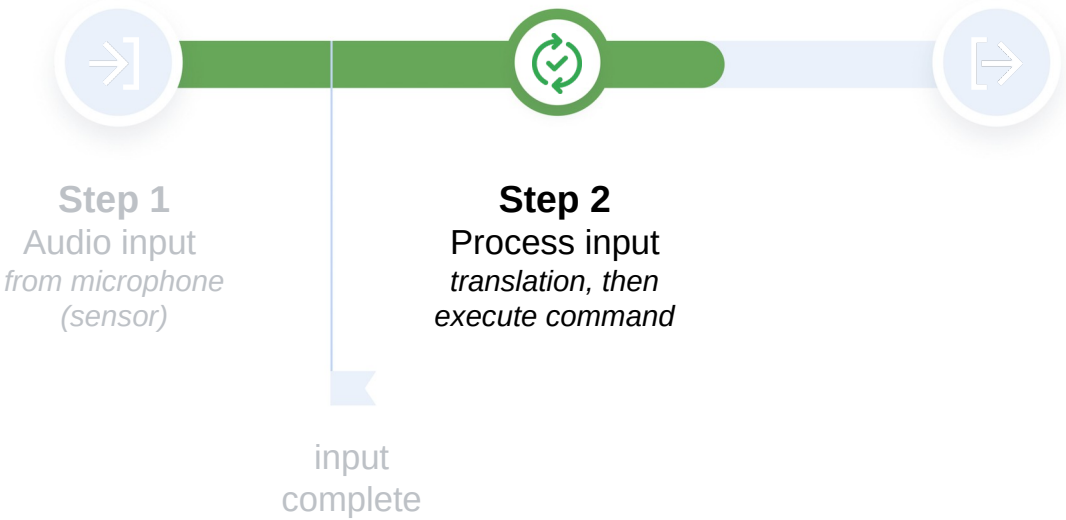


The Three Basic Steps

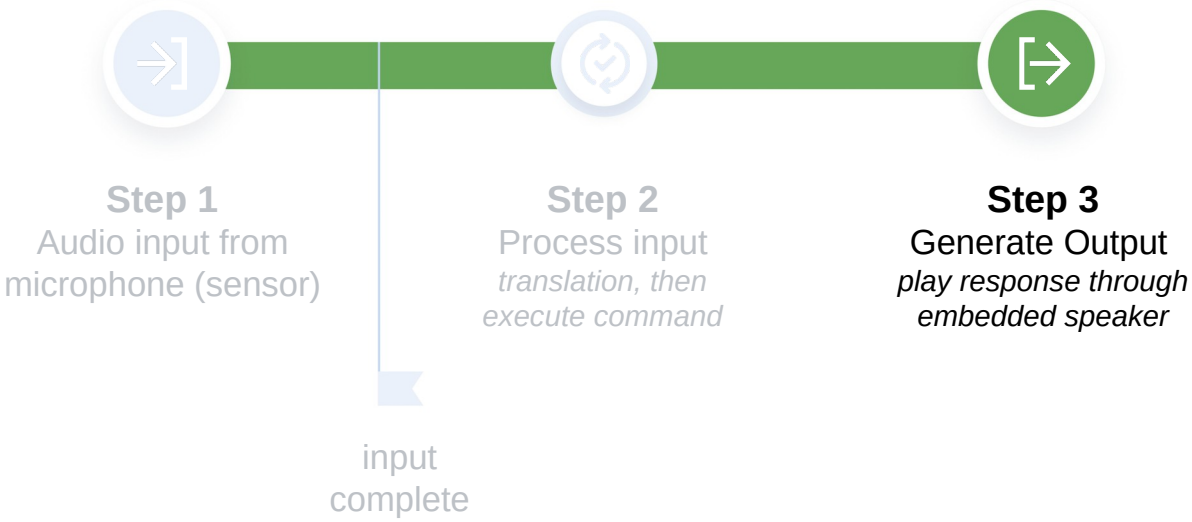
The Three Basic Steps



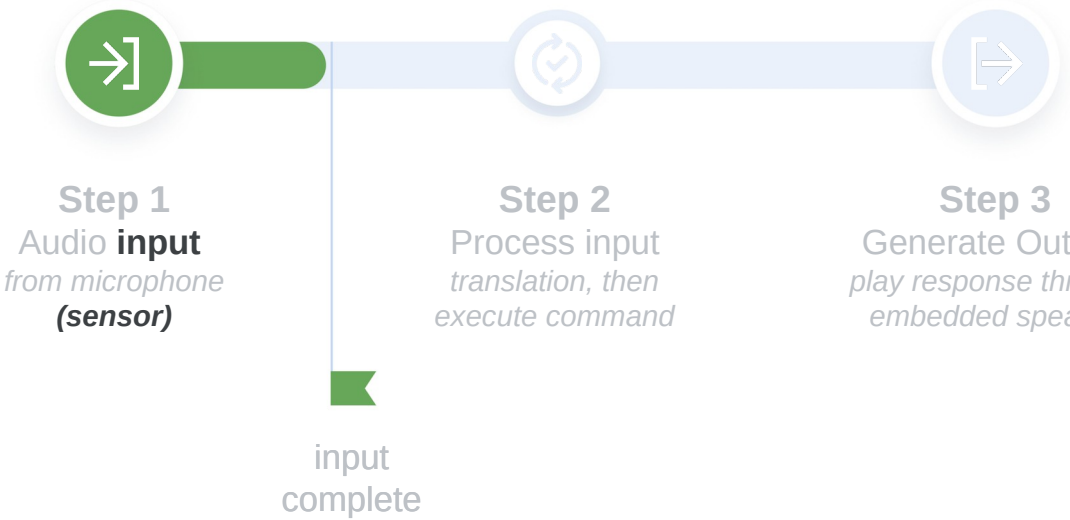
The Three Basic Steps



The Three Basic Steps



Input


Step 1
Audio **input**
from microphone
(sensor)
Step 2
Process input
translation, then
execute command
Step 3
Generate Output
play response through
embedded speaker
input
complete

Endpoints Have **Sensors**, Tons of Sensors

Motion Sensors
Gyroscope, radar, magnetometer, accelerator

Acoustic Sensors
Ultrasonic, Microphones, Geophones, Vibrometers

Environmental Sensors
Temperature, Humidity, Pressure, IR, etc.

Touchscreen Sensors
Capacitive, IR

Image Sensors
Thermal, Image

Biometric Sensors
Fingerprint, Heart rate, etc.

Force Sensors
Pressure, Strain

Rotation Sensors
Encoders

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Encoders

Biometric Sensors



Non-invasive Glucose Monitoring



Fingerprint + Photoplethysmography (**PPG**)

Source: Jacobs School of Engineering/UC San Diego

Endpoints Have **Sensors**, Tons of Sensors

Motion Sensors
Gyroscope, radar, magnetometer, accelerator

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Ultrasonic, Microphones, Geophones, Vibrometers

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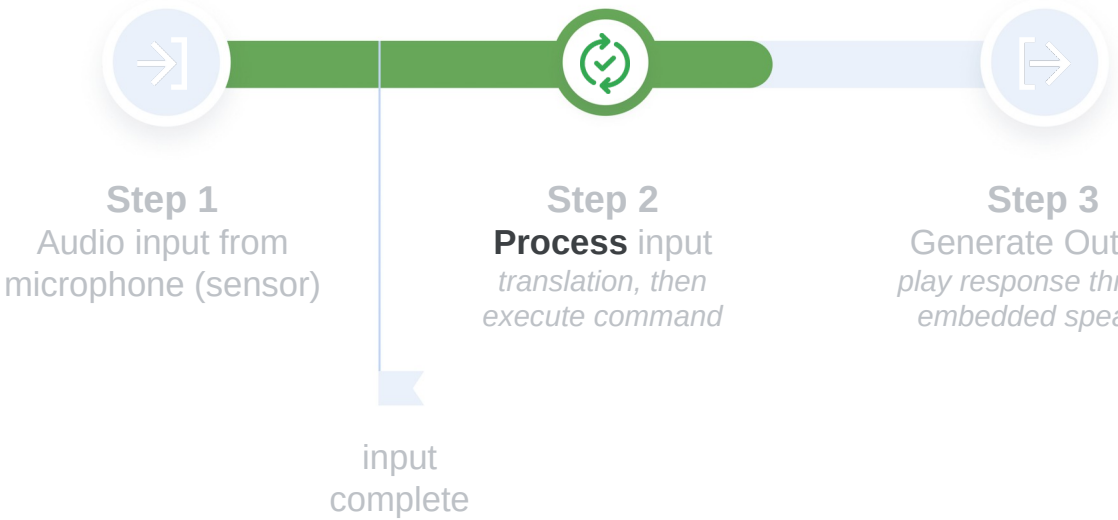
Force Sensors
Pressure, Strain

Rotation Sensors
Encoders

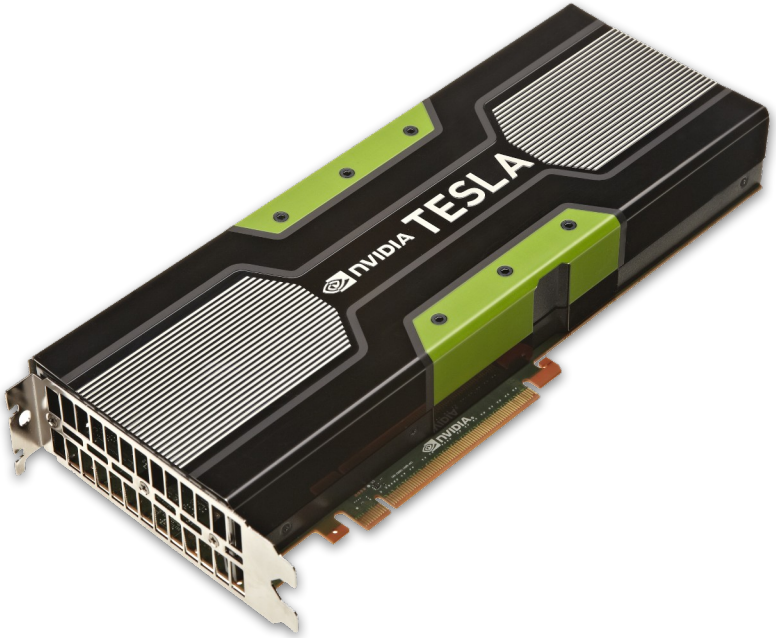
Endpoints Have **Sensors**, Tons of Sensors



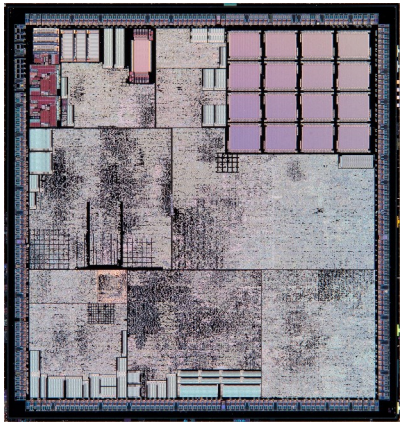
Processing


Step 1
Audio input from microphone (sensor)
Step 2
Process input
translation, then execute command
Step 3
Generate Output
play response through embedded speaker
input complete

Thinking **Big**



Thinking **Big**



Thinking **Big**

BIG

GPU / CPU

561mm²

Thinking **Small**

BIG

GPU / CPU

561mm²



Thinking **Small**

BIG

GPU / CPU

561mm²



Thinking **Small**





Mobile SoC
83mm²

Thinking **Tiny**

BIG

GPU / CPU

561mm²

SMALL

Mobile SoC

83mm²



Thinking **Tiny**

BIG
GPU / CPU
561mm²

SMALL

Mobile SoC
83mm²

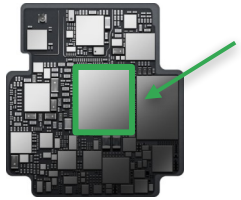


Thinking **Tiny**

BIG
GPU / CPU
561mm²

SMALL

Mobile SoC
83mm²



Thinking **Tiny**





Mobile SoC
83mm²



Apple 0778
30mm²

We’re just getting started.

Thinking **Record-breaking**





Mobile SoC
83mm²



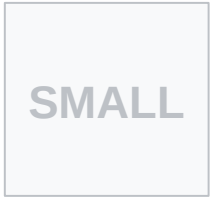
Apple 0778
30mm²



Kinetis KL03
3.2mm²

Thinking **Record-breaking**





Mobile SoC
83mm²



Apple 0778
30mm²

world's smallest ARM-Powered MCU

48MHz, 32KB flash, 20-pin



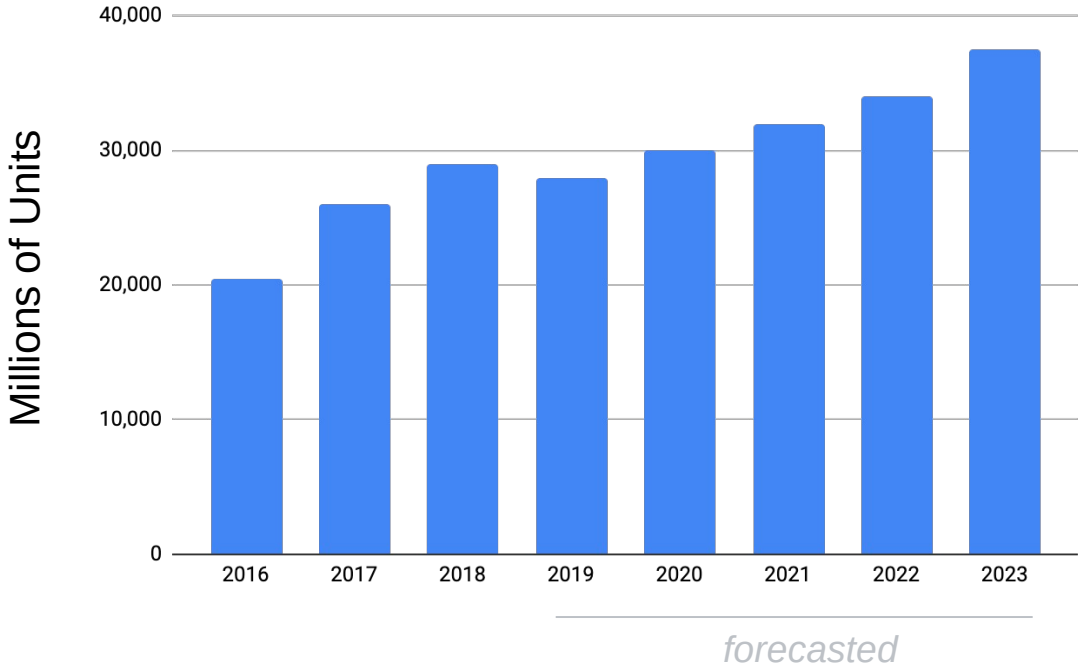
Kinetis KL03
3.2mm²


2.0 mm
1.6 mm

250 Billion

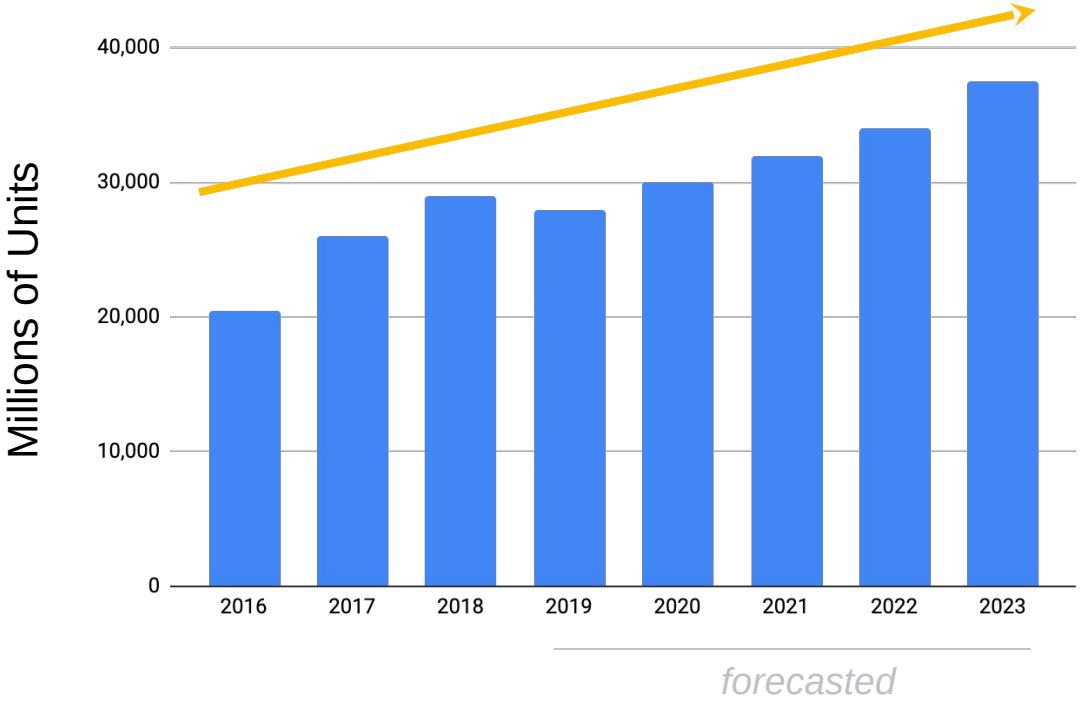
MCUs today

MCU Demand Forecast



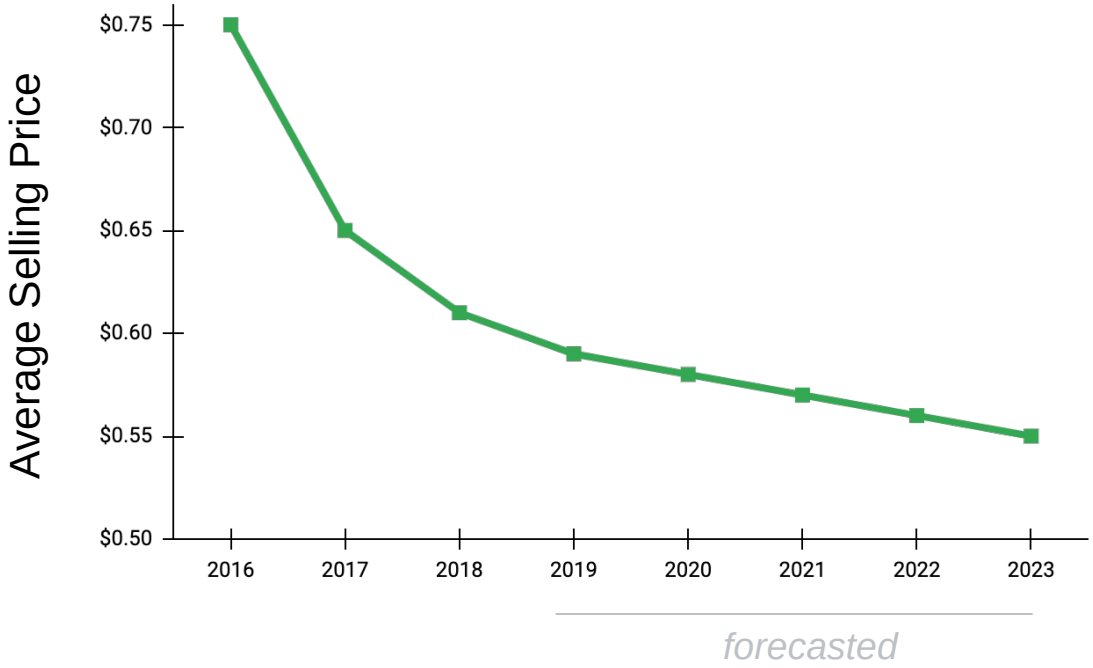
Source: IC Insights

MCU Demand Forecast



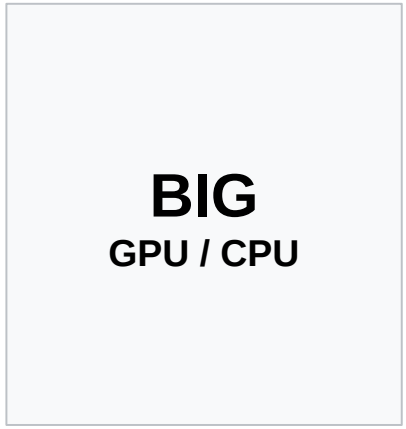
Source: IC Insights

MCU Pricing Forecast

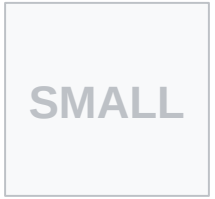


Source: IC Insights

Comparing Power



300W
NVIDIA Tesla K80

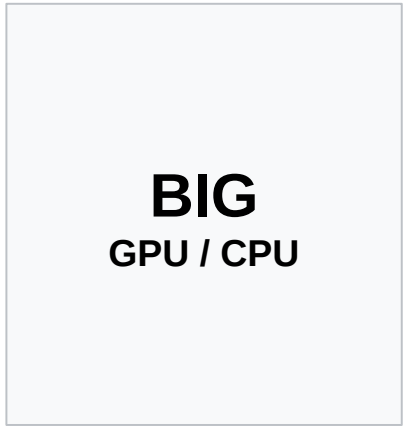


3.64W
Apple A12



140 µW
Syntiant NDP100

Comparing Power



300W
NVIDIA Tesla K80



3.64W
Apple A12



140 µW
Syntiant NDP100

Comparing Power

BIG
GPU / CPU

300W
NVIDIA Tesla K80

SMALL

3.64W
Apple A12

Neural Decision Processor
Always-on deep learning
speech/audio recognition

Ultra low power, 128KB
SRAM, 12-pin, 2.52mm^2



140 µW
Syntiant NDP100

Comparing Power



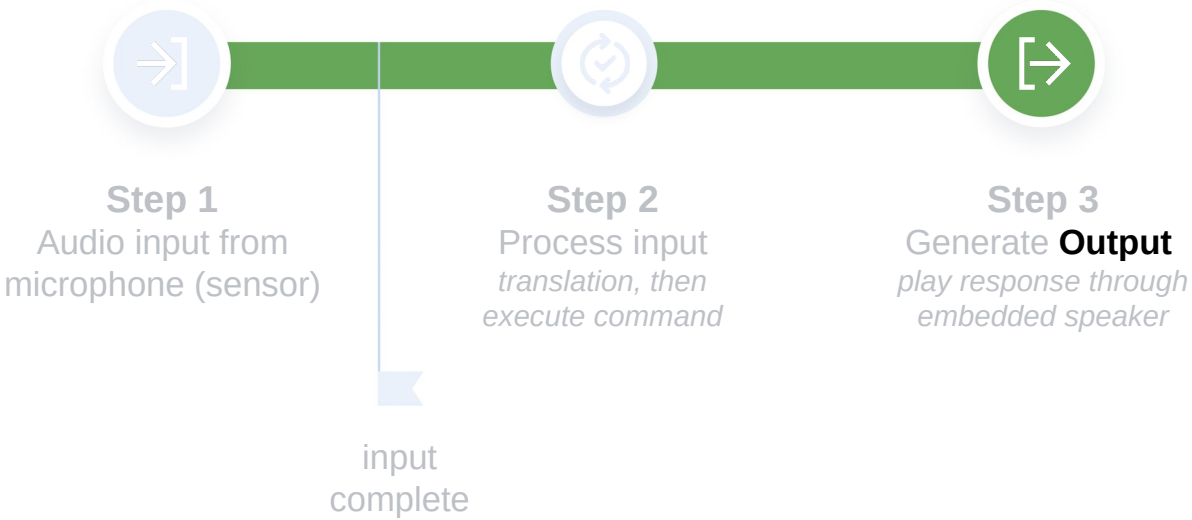
Use case: button cell battery

Neural Decision Processor
Always-on deep learning
speech/audio recognition

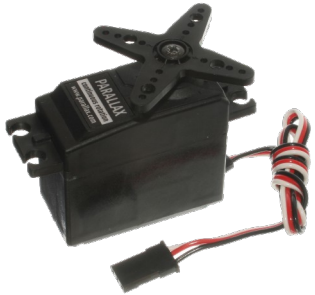
Ultra low power, 128KB
SRAM, 12-pin, 2.52mm^2

140 µW
Syntiant NDP100

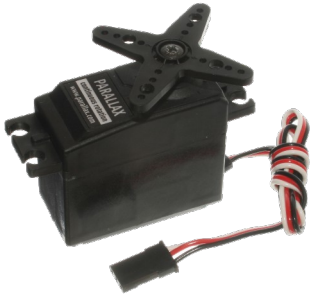
Output


Step 1
Audio input from microphone (sensor)
Step 2
Process input
translation, then execute command
Step 3
Generate Output
play response through embedded speaker
input complete

Output

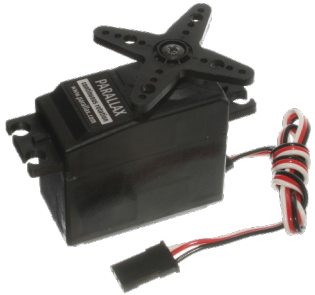


Output





Output


PARALLAX




InVision
8:00 PM
100%
Properties
103 FIF
103 Fife Street
Nashville, TN 37206
Noise Risk Score
30 minutes
Alert!
100
90
80
70
60
50
40
30
20
10
0
9:35pm PST
10:05pm PST
Living Room
Hot Tub
Patio
Data
Sensors
Events
Settings

MCUs enable **TinyML**

SIZE

LOW POWER

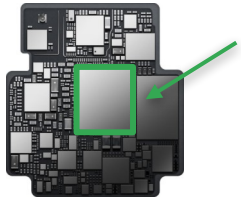
LOW COST

MCUs enable **TinyML**

SIZE

LOW POWER

LOW COST



MCUs enable **TinyML**

SIZE

LOW POWER

LOW COST

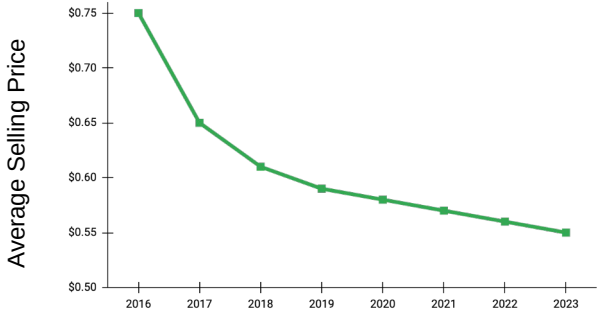


< 140 µW

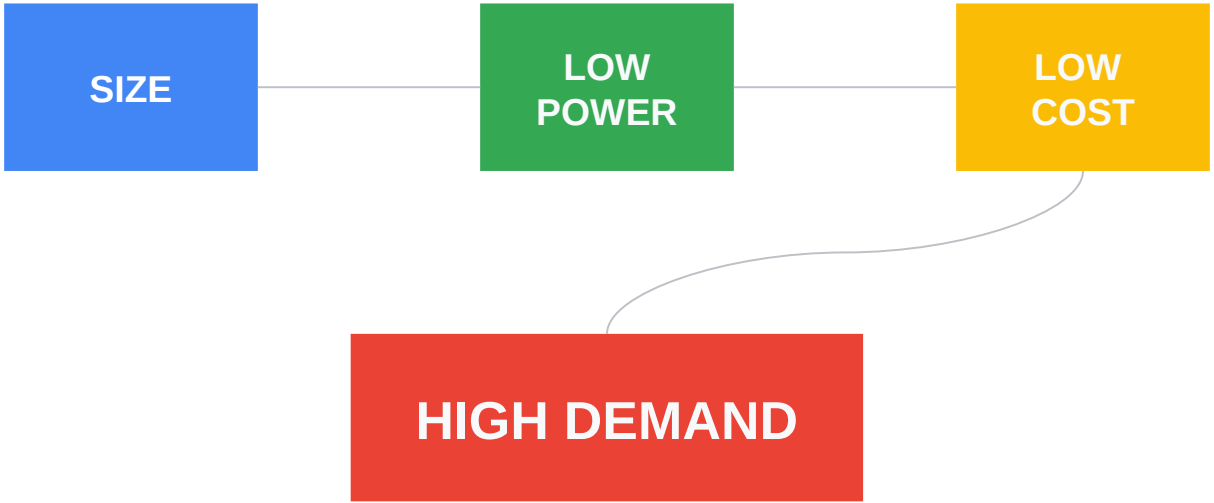
Syntiant NDP100

MCUs enable **TinyML**





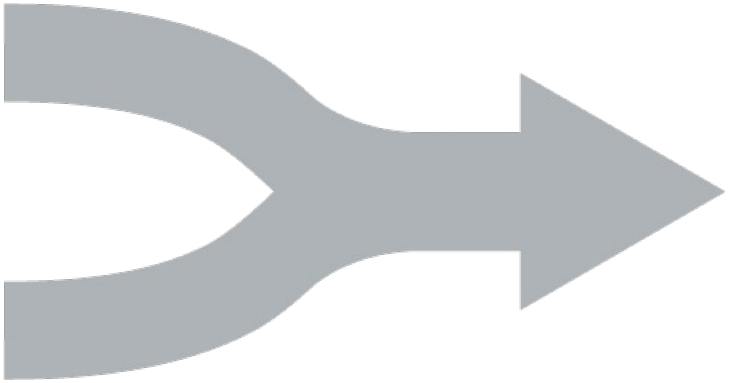
MCUs enable **TinyML**



What Makes **TinyML**?

Embedded Systems

Machine Learning

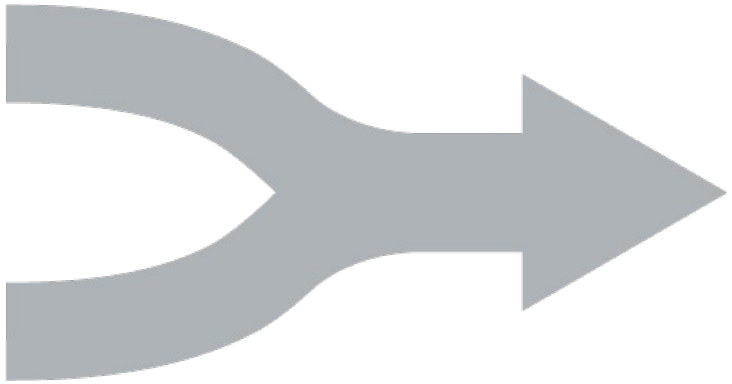


TinyML

What Makes **TinyML**?

Embedded Systems

Machine Learning

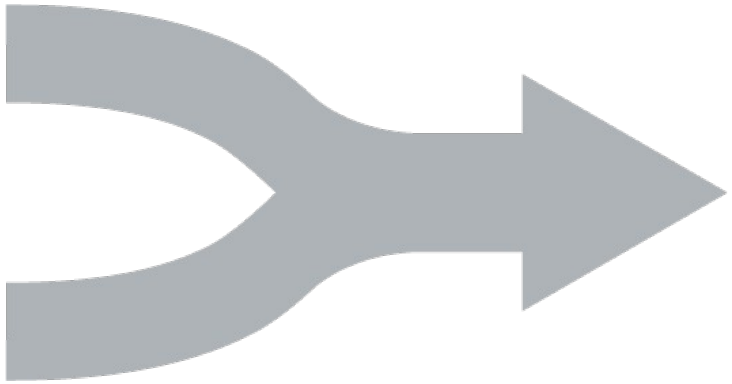


TinyML

What Makes **TinyML**?

Embedded Systems

Machine Learning



TinyML

Fullscreen. Show presenter.